

OSM / Opto-PING Rung 0L

Standard-PING Null and Empty-Sky Specificity Gate v0.1

Overall status: PARTIAL_OR_FAILED_SPECIFICITY

Rung 0L asks the missing false-positive question: if the synthetic sky is empty - no reciprocal hidden Y-loop and true K = 0 - does the Opto-PING full reciprocal model stay quiet, or does it hallucinate a loop?

This gate complements Rung 0J/K. Rung 0J/K tested positive recovery when the reciprocal loop exists. Rung 0L tests specificity when Standard-PING, replay-only, sensory-drive-only, generic hidden oscillator, or colored noise generators create the data without an Opto-PING loop.

Rung 0L spec SHA-256: 07b72130147812bd9f4cbf206bc224ff2af05874ab9ad1f554ed80310a64e8ae

Boundary: Synthetic specificity only; no biological, PR-AYC-G, human EEG, or OSM mechanism claim.

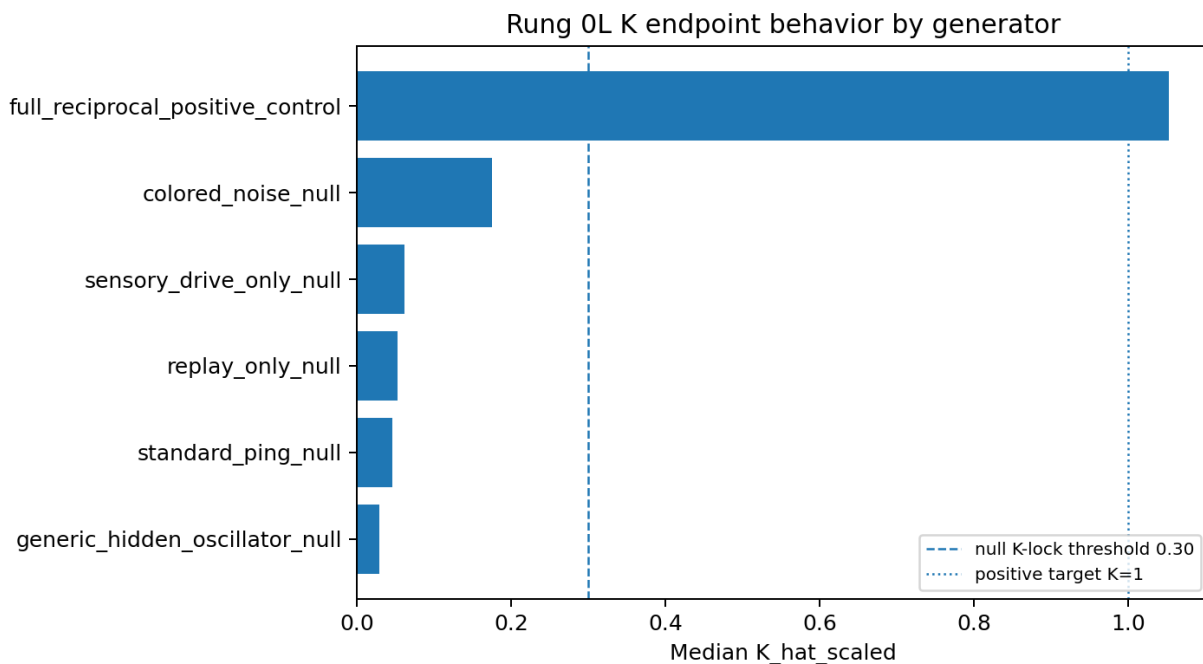
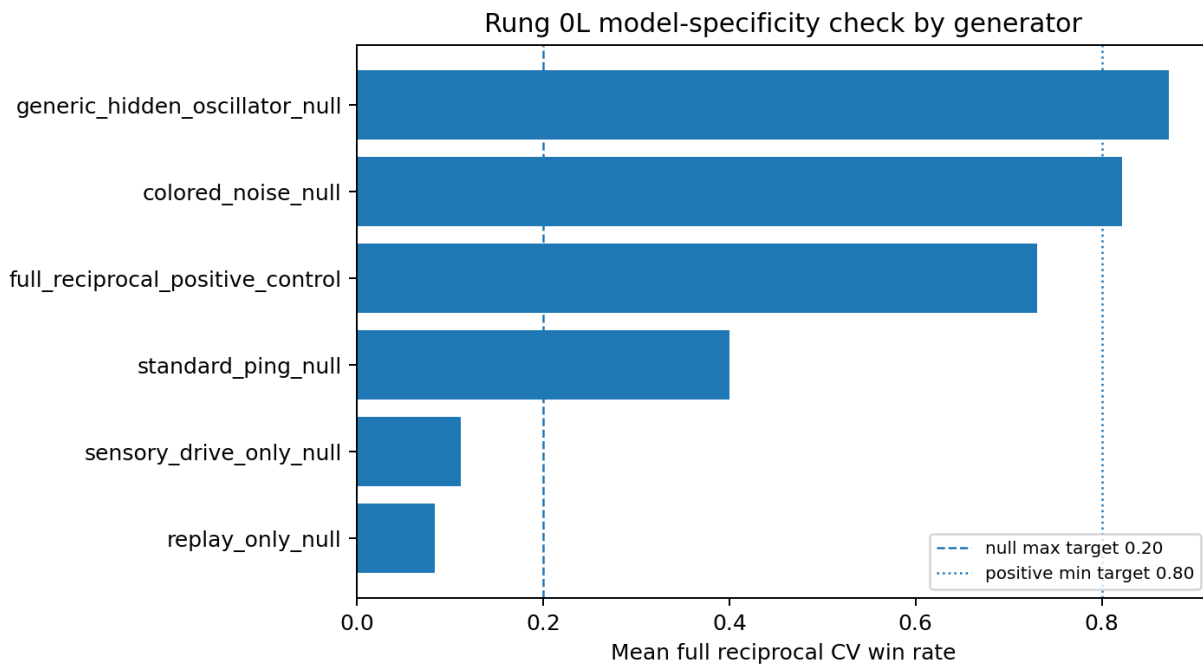
1. Main numerical result

Criterion	Observed	Threshold	Pass
Positive control: full reciprocal CV win rate	0.731	≥ 0.80	False
Positive control: K within 15% rate	1.000	≥ 0.80	True
Empty-sky nulls: full reciprocal CV win rate	0.458	≤ 0.20	False
Empty-sky nulls: false K-lock rate	0.023	≤ 0.10	True
Empty-sky nulls: non-full CV win rate	0.542	≥ 0.80	False
Empty-sky nulls: median $K_{\text{hat_scaled}}$	0.054	≤ 0.30	True

The primary finding is mixed. The K endpoint behaved well: the positive generator recovered K and the null generators had low median K and low overall false K-lock. However, the full reciprocal model won too often in held-out CV on several null generators. Therefore Rung 0L is not a clean specificity pass.

2. Generator-level results

generator	n	K_true	median K	false K-lock	full CV	non-full CV
colored_noise_null	60	0.0	0.175	0.117	0.822	0.178
full_reciprocal_positive_control	60	1.0	1.053	0.000	0.731	0.269
generic_hidden_oscillator_null	60	0.0	0.029	0.000	0.872	0.128
replay_only_null	60	0.0	0.053	0.000	0.083	0.917
sensory_drive_only_null	60	0.0	0.062	0.000	0.111	0.889
standard_ping_null	60	0.0	0.045	0.000	0.400	0.600



3. What passed

The positive-control generator recovered the composite K endpoint: K within 15% was 1.000 and K within 10% was 0.933, with median $K_{\hat{scaled}} = 1.053$. This says the K estimator still detects the loop when the loop is present.

Across all empty-sky null generators, median $K_{\hat{scaled}}$ was only 0.054 and the overall false K-lock rate was 0.023. That is a useful result: the scalar K endpoint did not broadly hallucinate a strong loop under $K_{\text{true}} = 0$.

4. What failed or weakened

The full reciprocal model did not stay quiet in model comparison. Its mean CV win rate across null generators was 0.458, above the ≤ 0.20 specificity target. In particular, it won 0.872 of folds on the generic-hidden-oscillator null and 0.822 on the colored-noise null. This means the current full-model feature space is too permissive under some

structured nulls.

The Standard-PING null was only partially controlled: median K stayed near zero, but the full model still won 0.400 of held-out folds. Replay-only and sensory-drive-only nulls behaved better: full model win rates were 0.083 and 0.111, respectively.

5. Interpretation

Rung 0L does not invalidate Rung 0J/K. Rung 0J/K remains a positive synthetic recovery result. Rung 0L adds the missing specificity stress test and shows that the K endpoint is comparatively well-behaved, but the model-selection layer needs stronger null-aware constraints before the computational ladder can claim robust empty-sky specificity.

The most important conclusion is methodological: the radar does not strongly hallucinate K, but the current full reciprocal model can still over-explain structured null skies. The next gate should improve the null alternatives and complexity penalties rather than celebrate prematurely.

6. Recommended next gate - Rung 0M

Rung 0M should be a conservative null-aware model-selection gate. It should keep the Rung 0L generators, but add generator-specific pass/fail thresholds, BIC/AIC or nested-CV penalties, stronger colored-noise and generic-hidden alternatives, permutation/null calibration of K, and a rule that the full reciprocal model must win both prediction and K-lock simultaneously. K-lock alone should not be enough; full-model prediction alone should not be enough.

7. Files generated

The package includes the locked Rung 0L spec, trial-level results, CV model losses, fold-winner tables, generator summaries, lock-decision table, figures, this PDF report, and the reproducibility script.